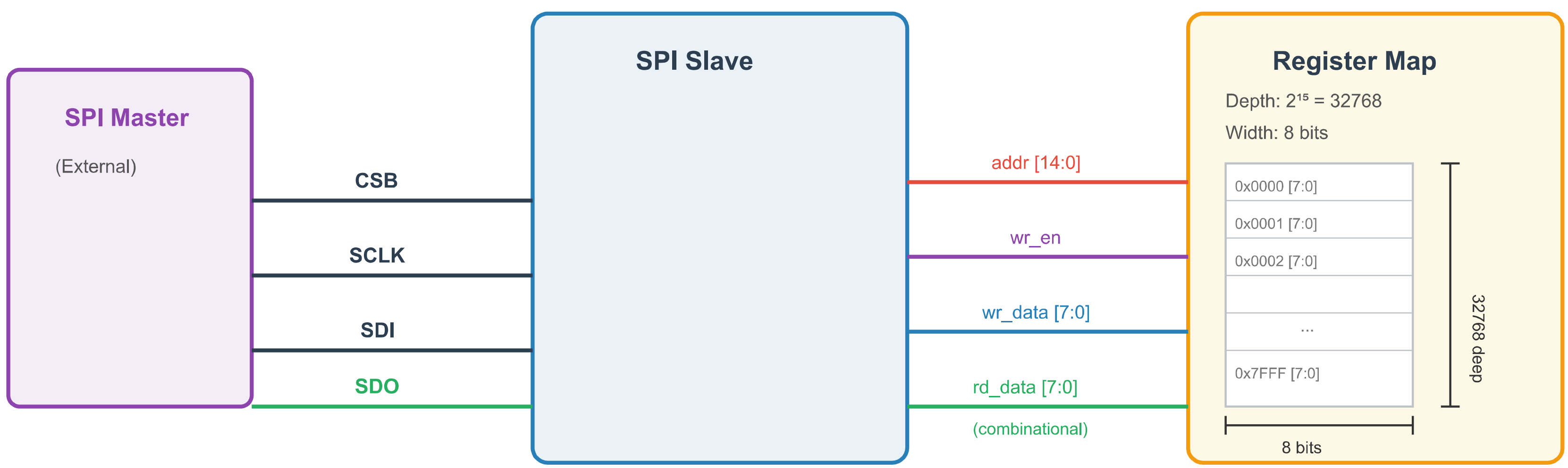
Assignment: 4-Wire SPI Slave Interface

1. Block Diagram



2. Objective

Design a standard 4-wire SPI slave controller (CSB, CLK, SDI, SDO) that provides read/write access to an internal register map.

3. Frame Format

Each SPI transaction consists of a **16-bit header** followed by one or more **8-bit data bytes**. All bits are transmitted MSB-first.

```
Single Byte Access (24 bits):

┌──────────────────────────────────────────────┬─────────────────────────────┐
│            HEADER (16 bits)                  │        DATA (8 bits)        │
├─────┬────────────────────────────────────────┼─────────────────────────────┤
│ R/W │           ADDRESS [14:0]               │           D[7:0]            │
├─────┼────────────────────────────────────────┼─────────────────────────────┤
│ b15 │ b14  b13  b12  ...  b3  b2  b1  b0 │ b7  b6  b5  ...  b1  b0 │
└─────┴────────────────────────────────────────┴─────────────────────────────┘
◄──────────────── 16 clocks ──────────────►◄─────── 8 clocks ───────►


Multi-Byte Burst (auto-increment):

┌───────────────────────────────────────────┬────────┬────────┬────────┬─────┐
│          HEADER (16 bits)                 │ BYTE 0 │ BYTE 1 │ BYTE 2 │ ... │
├─────┬─────────────────────────────────────┼────────┼────────┼────────┼─────┤
│ R/W │        ADDRESS [14:0]               │ D[7:0] │ D[7:0] │ D[7:0] │     │
└─────┴─────────────────────────────────────┴────────┴────────┴────────┴─────┘
                                             addr+0   addr+1   addr+2
```

4. Requirements

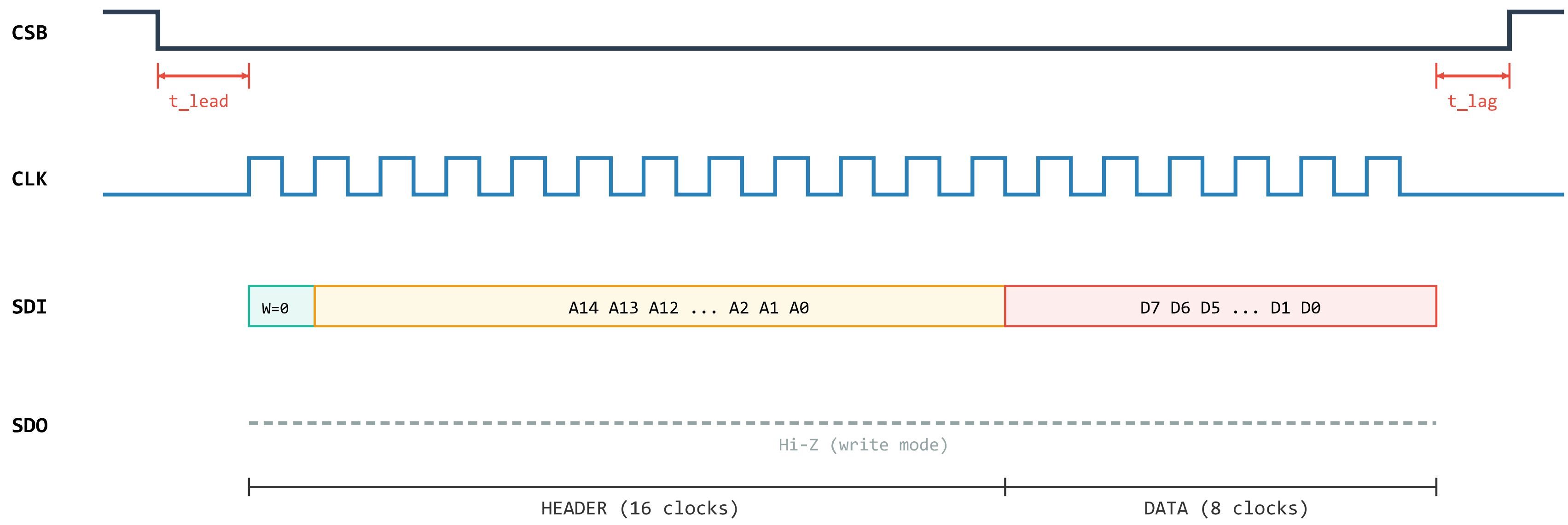
- **R/W Bit (bit[15]):** 0 = Write, 1 = Read.

- **Address (bit[14:0]):** 15-bit register address, targeting the internal register map.
- **Data:** Minimum one byte (8 bits). Additional consecutive bytes indicate burst (auto-increment) mode.
- **Increment Mode:** There is no dedicated bit in the frame to signal burst mode. The slave detects increment mode by observing continued CLK activity and data after the first byte completes. The address auto-increments by 1 for each subsequent byte.
- **Register Map Read Timing:** The register map interface must be **combinationally readable**. On a read transaction, the slave has only **half a CLK cycle** to prepare the first data bit (MSB of D[7:0]) after the last address bit (LSB of A[14:0]) is clocked in. The read data must appear on SDO at the very next CLK edge.
- **CSB Timing Margins:**
  - **t_lead:** A predefined delay must exist between CSB assertion (falling edge) and the first CLK rising edge.
  - **t_lag:** A predefined delay must exist between the last CLK falling edge and CSB de-assertion (rising edge).
  - You may assume **t_lag = t_lead**.

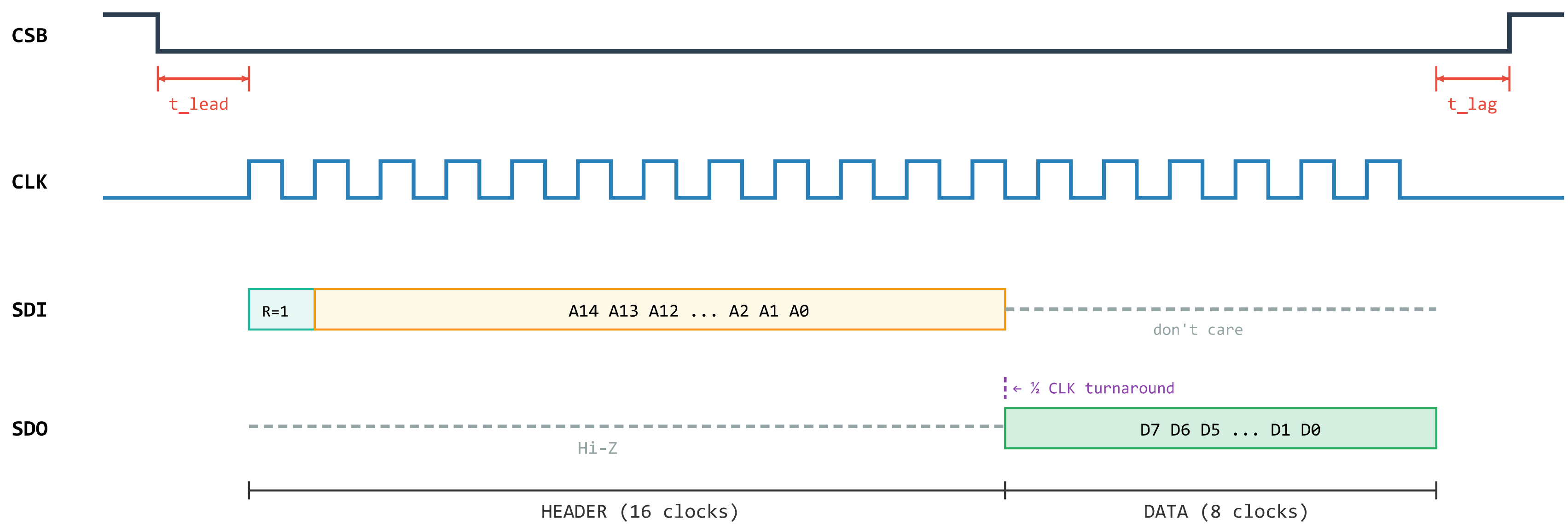
5. Field Definitions

Field	Bits	Description
R/W	[15]	0 = Write (master sends data on SDI) 1 = Read (slave drives data on SDO)
ADDRESS	[14:0]	Starting register address (auto-increments in burst)
DATA	8 per byte	Register data, MSB first

6. Timing Diagram — Write Transaction



7. Timing Diagram — Read Transaction



Read Turnaround: After the last address bit (A0) is sampled on the rising edge of CLK cycle 16, the slave must drive the MSB of read data (D7) on SDO before the rising edge of CLK cycle 17. This gives the slave only **half a CLK period** to read the register map and present data — hence the requirement for a combinational read path.

8. Burst Behavior

- CSB remains asserted (low) beyond the initial 24 clocks to signal a burst.
- No explicit bit in the header indicates burst mode — it is detected by continued CLK activity after the first data byte.
- The internal address register increments by 1 after each complete byte.
- The burst terminates when CSB de-asserts (rising edge), respecting the t_lag timing constraint.